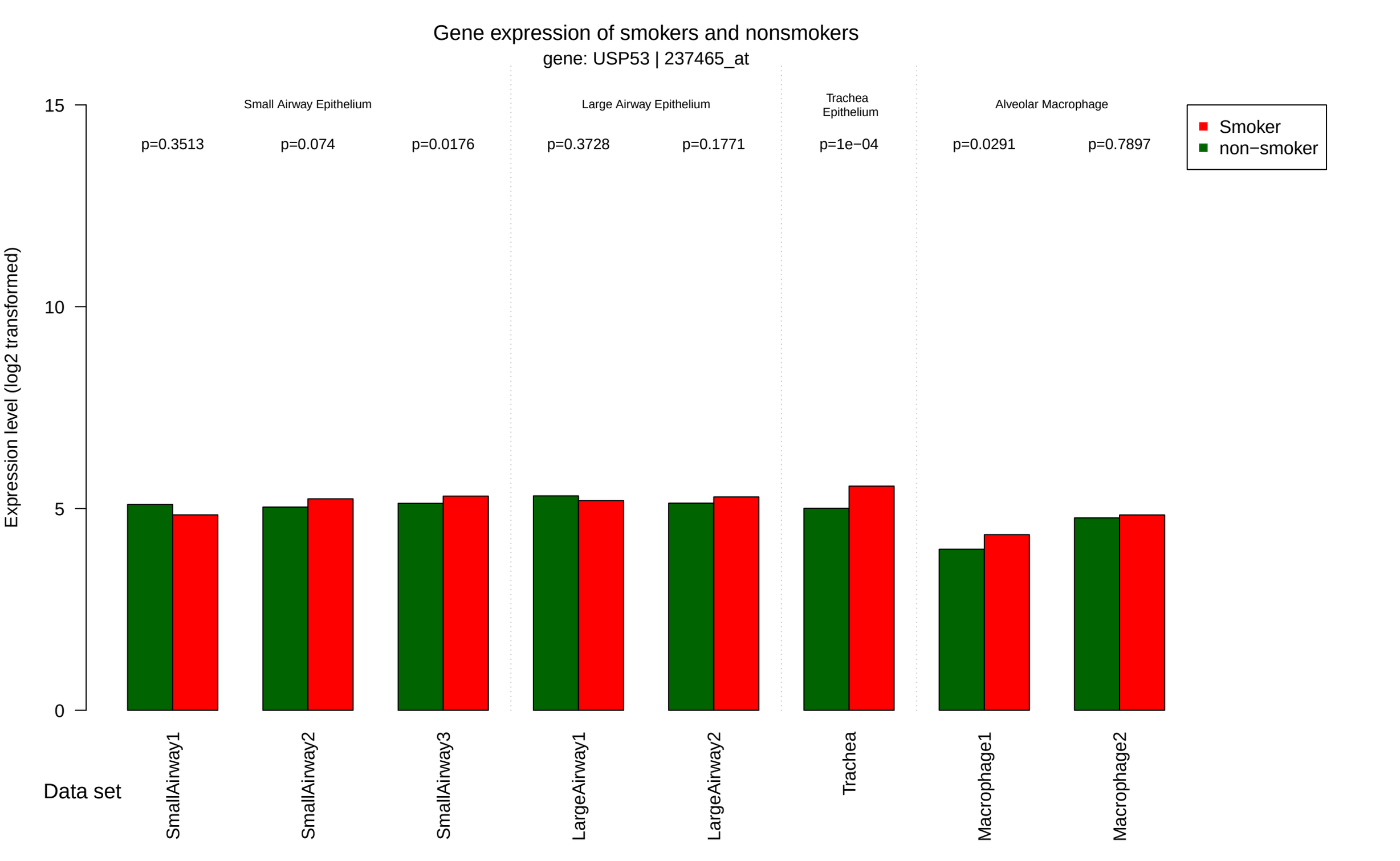
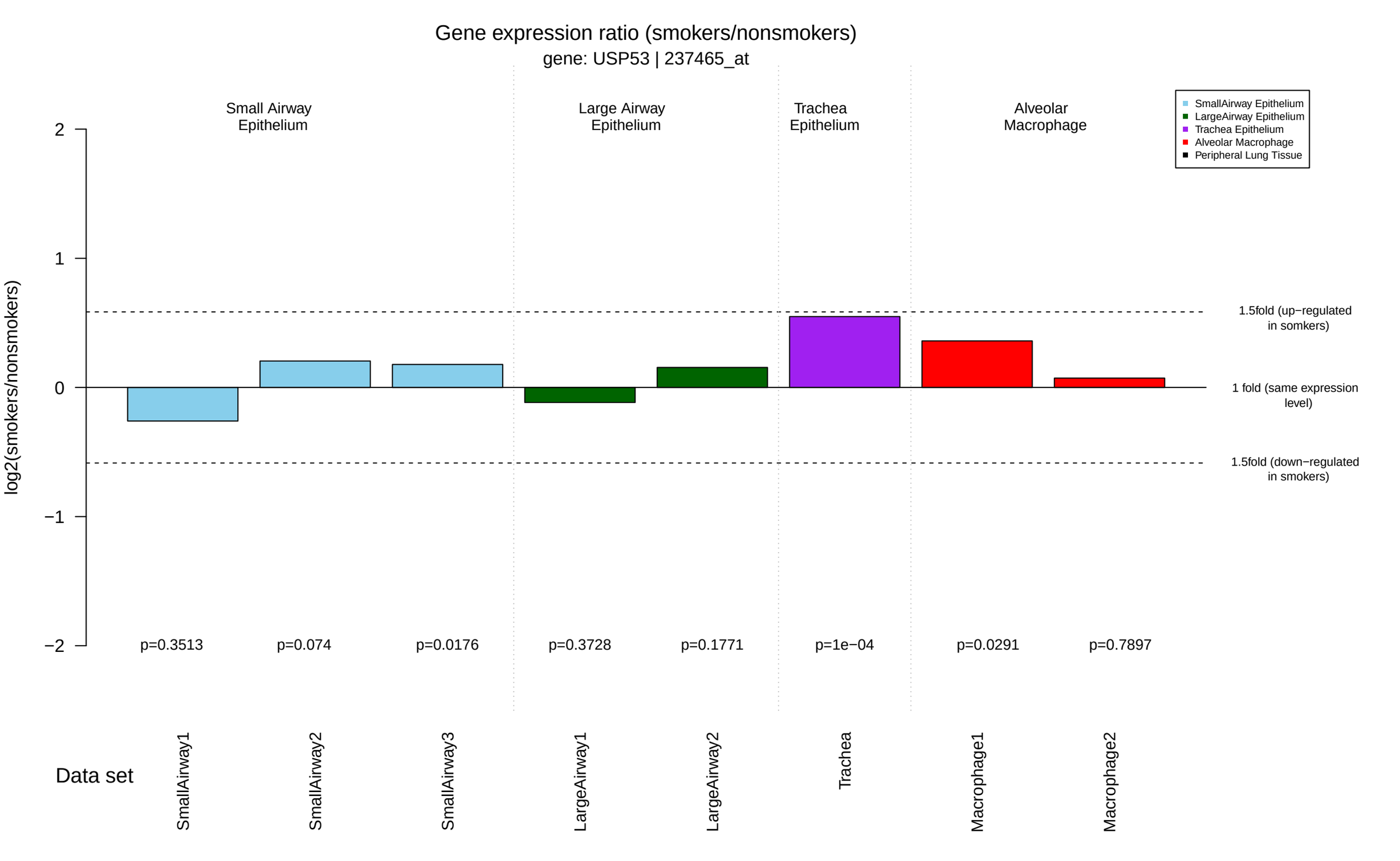
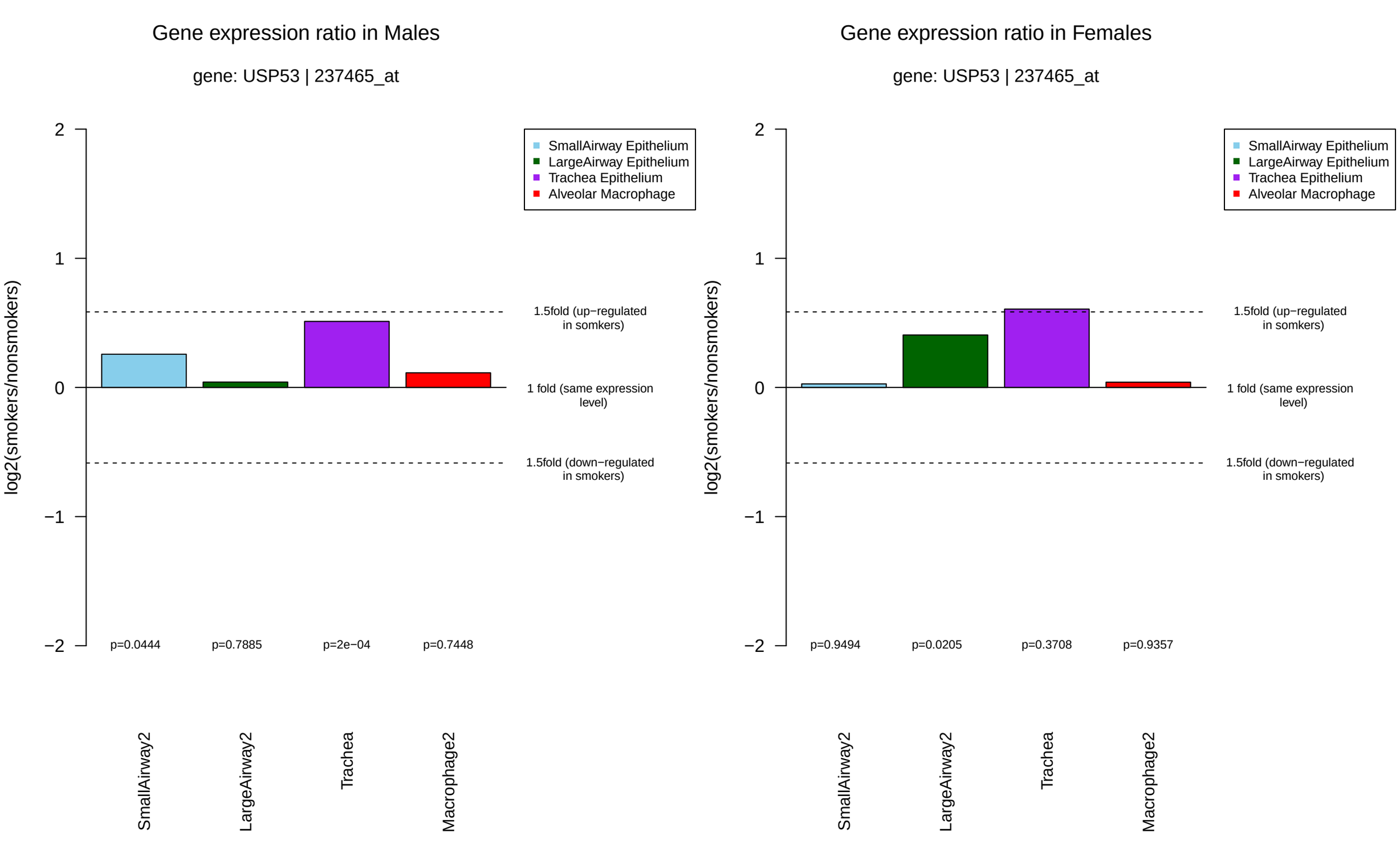
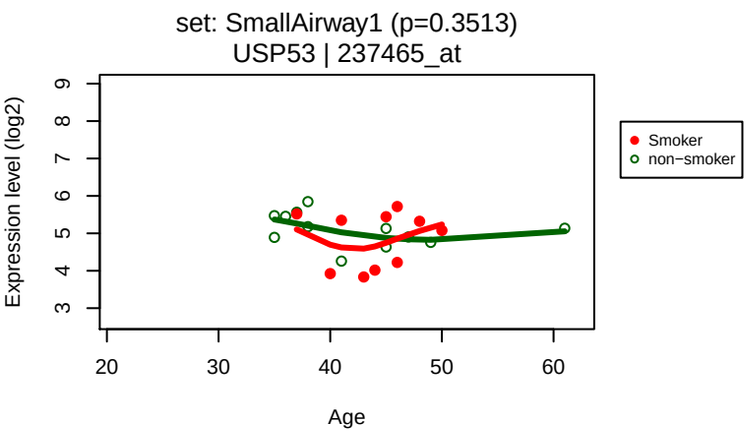
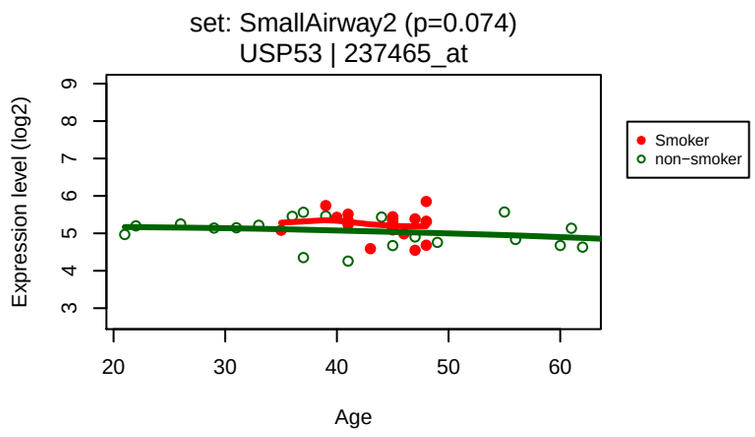
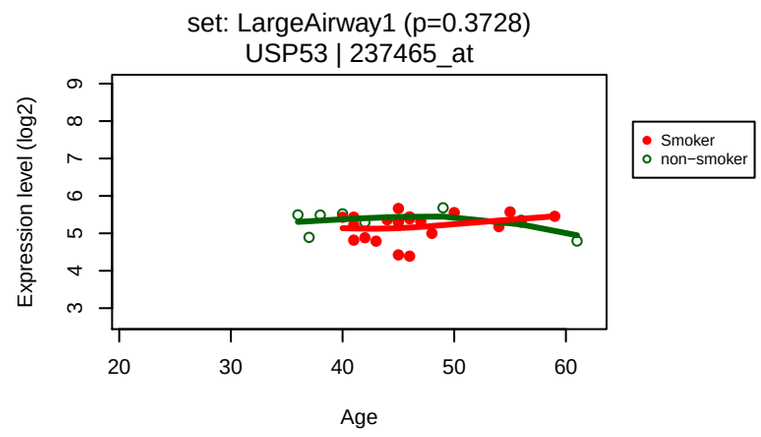

Gene expression of smokers and nonsmokers
gene: USP53 | 237465_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3513
p=0.074
p=0.0176
p=0.3728
p=0.1771
p=1e−04
p=0.0291
p=0.7897
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

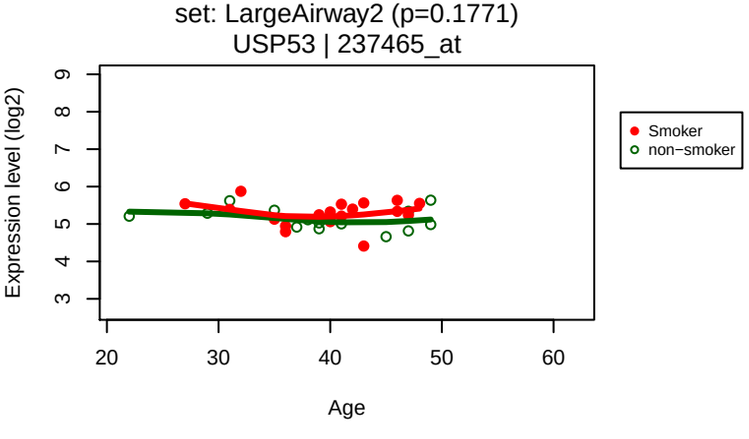

Gene expression ratio (smokers/nonsmokers)
gene: USP53 | 237465_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3513
p=0.074
p=0.0176
p=0.3728
p=0.1771
p=1e−04
p=0.0291
p=0.7897
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

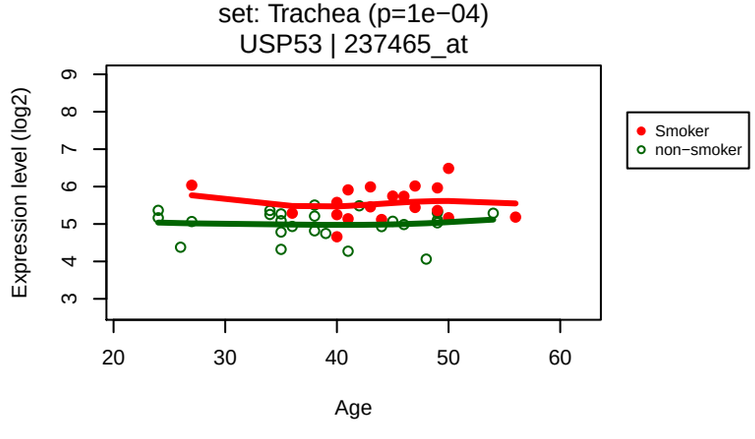

Gene expression ratio in Males
gene: USP53 | 237465_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0444
p=0.7885
p=2e−04
p=0.7448
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: USP53 | 237465_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9494
p=0.0205
p=0.3708
p=0.9357
SmallAirway2
LargeAirway2
Trachea
Macrophage2

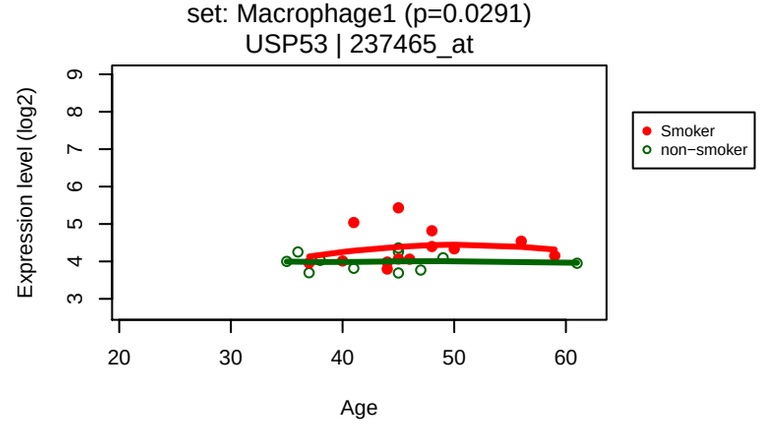

set: SmallAirway1 (p=0.3513)
USP53 | 237465_at
Expression level (log2)
Age
Smoker
non−smoker

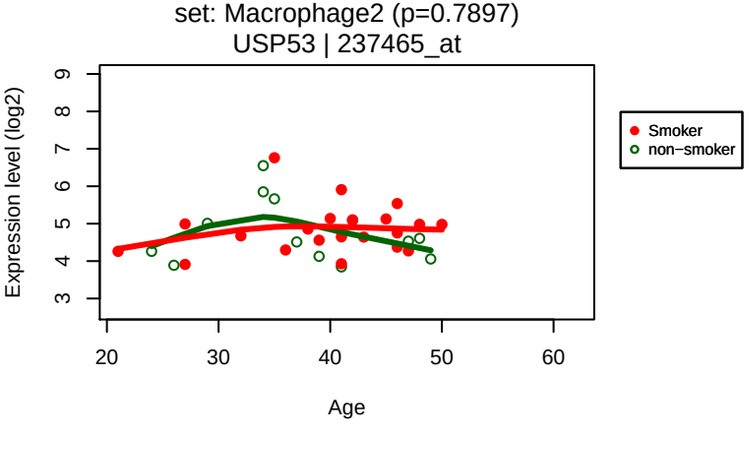

set: SmallAirway2 (p=0.074)
USP53 | 237465_at
Expression level (log2)
Age
Smoker
non−smoker

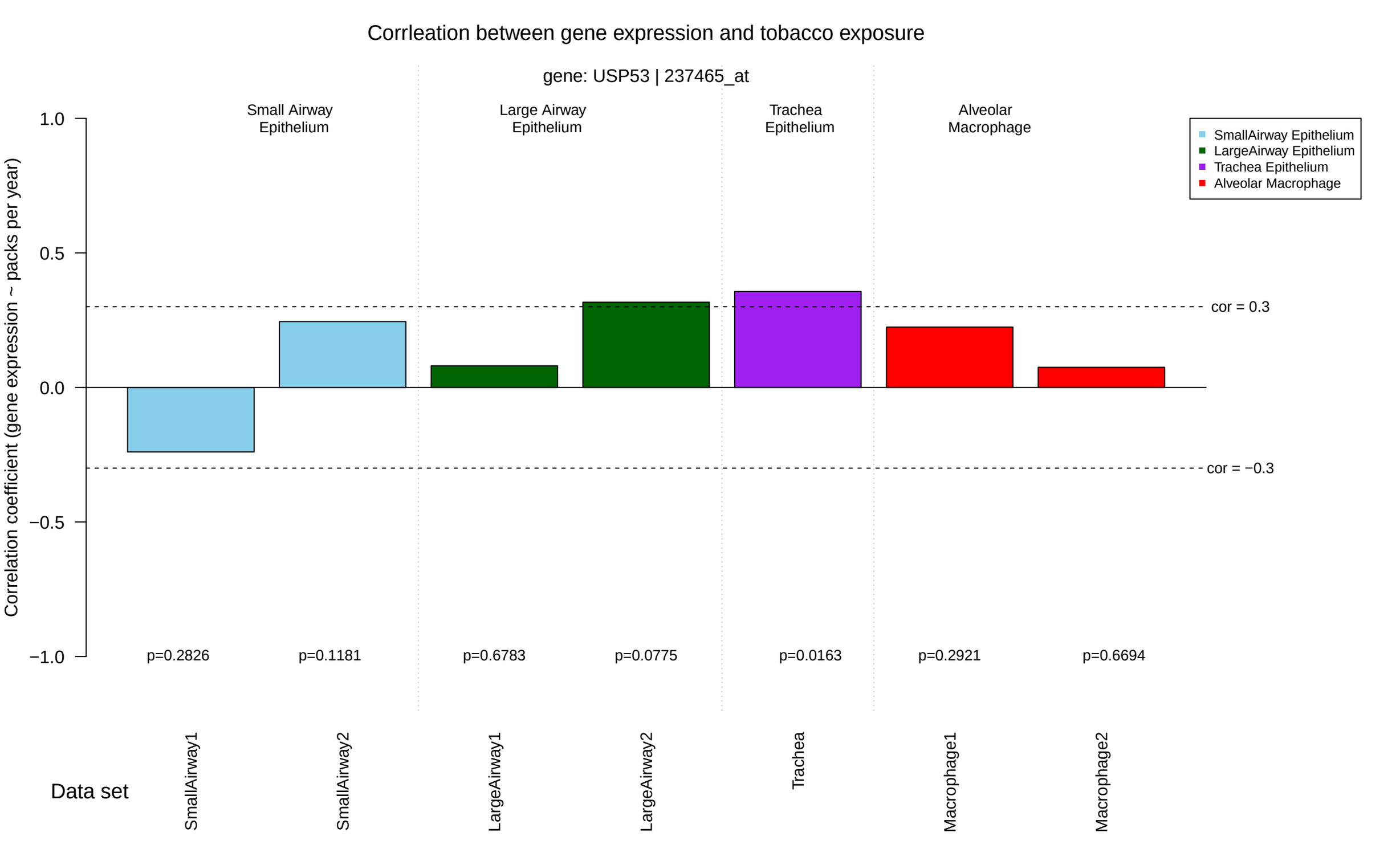

set: LargeAirway1 (p=0.3728)
USP53 | 237465_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1771)
USP53 | 237465_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=1e−04)
USP53 | 237465_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0291)
USP53 | 237465_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7897)
USP53 | 237465_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: USP53 | 237465_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2826
p=0.1181
p=0.6783
p=0.0775
p=0.0163
p=0.2921
p=0.6694
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2